

Sangyun Lee

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Work Experience

Sungkonghoe University

Role **Date** Sep.2021 – Present

- Assistant Professor in Division of Future Convergence

Projects

- A Study on Vision-Language-based Multi-Object Tracking (Sungkonghoe University, Sep.2025 – Aug.2027)
- Making Action Detection Practical: Leveraging 2D CNNs for Real-Time and Online Applications (Sungkonghoe University, Aug.2024 – Jul.2025)
- Real-time monitoring technology for identifying hazardous situations for workers in industrial settings using video-based AI (Ministry of SMEs and Startups, Jul.2024 – Dec.2026)
- Development of Core Technology for Preventing Identity Theft (SecuChart Global, May.2023 – Dec.2023)
- A Context-aware Multiple Object Tracking with Temporal Shifted Transformer (Sungkonghoe University, Aug.2022 – Jul.2023)
- Research on AI integration technology for comprehensive context-aware multiple object tracking (Ministry of Science and ICT, Mar.2022 – Feb.2024)

Robot Center, Samsung Research, Samsung Electronics Co., Ltd.

Role **Date** Mar.2019 – Aug.2021

- Develop on-device AI models for robot perception
- Design SW architecture for human-robot interaction

Autonomous Machine Lab, Samsung Research, Samsung Electronics Co., Ltd.

Role **Date** Mar.2018 – Feb.2019

- Develop deep learning models for autonomous driving
- Develop multiple object tracking system with sensor fusion

Education

Yonsei University

Date Sep.2011 – Feb.2018

- Ph.D. in Electrical & Electronic Engineering
- Advisor: Prof. Euntai Kim (etkim@yonsei.ac.kr)
- Dissertation "Deep Neural Network based Discriminative Feature Learning and Its Application to Multiple Object Tracking"

Yonsei University

Date Mar.2007 – Aug.2011

- B.S. in Electrical & Electronic Engineering

Publications (main author only)

Journals

- S. Chun, S. Song, D. Lee, **S. Lee**, "StoneGAT: A Robust Fall Detection Framework via Skeleton-aware Graph Attention Networks," *International Journal of Control, Automation and Systems*, vol. 23, no. 12, pp. 3702-3713, 2025.
- **S. Lee**, "Feature Refinement With Vision State Space Modules for Tiny Object Detection," *Journal of Institute of Control, Robotics and Systems*, vol. 31, no. 7, pp. 797-801, 2025.
- M. Park, Y. A. Kim, H. Cha and **S. Lee**, "DeepCampus: A Campus Tour Mobile Application Using Deep Learning-Based Vision Technologies," *The Transactions of the Korean Institute of Electrical Engineers*, vol. 74, no. 1, pp. 127-134, Jan, 2025.
- S. Ryu, S. Hong and **S. Lee**, "Making TSM Better: Preserving Foundational Philosophy for Efficient Action Recognition," *ICT Express*, vol. 10, no. 3, pp. 570-575, Jun, 2024.
- **S. Lee** and S. Hong, "Extended Siamese Convolutional Neural Networks for Discriminative Feature Learning," *International Journal of Fuzzy Logic and Intelligent systems*, vol. 22, no. 4, pp. 339-349, Dec, 2022.
- **S. Lee** and E. Kim, "Multiple Object Tracking via Feature Pyramid Siamese Networks," *IEEE Access*, vol. 7, pp. 8181-8194, Dec, 2019.
- **S. Lee**, M. Park and J. Baek, "Robust adaptive synchronization of a class of chaotic systems via fuzzy bilinear observer using projection operator," *Information Sciences*, vol. 402, pp. 182-198, Sep, 2017.
- **S. Lee**, S. Jang, M. Park and C. Yoon, "Circuit Modeling and Simulation for Thermoelectric Cooling System using Condensed Water," *Journal of Korean Institute of Intelligent Systems*, vol. 25, no. 2, pp. 161-167, Apr. 2015.
- **S. Lee**, H. Seo, C. Hyun and M. Park, "Fuzzy Disturbance Observer based Multiple Sliding Surface Control of Nonlinear Systems with Mismatched Disturbance," *Journal of Korean Institute of Intelligent Systems*, vol. 24, no. 4, pp. 385-391, Aug. 2014.
- **S. Lee**, M. Park, and J. Baek, "Modeling of Dynamic Hysteresis Based on Takagi-Sugeno Fuzzy Duhem Model," *International Journal of Fuzzy Logic and Intelligent Systems*, vol. 13, no. 4, pp. 277-283, Dec. 2013.

Proceedings

- S. Chun, S. Song, D. Lee, S. Lee, "StoneGAT: A Robust Fall Detection Framework via Skeleton-aware Graph Attention Networks," in *Proc. of The 25th International Conference on Control, Automation and Systems (ICCAS 2025)*, Incheon, Korea, Nov, 2025.
- S. W. Kwak, S. Hong and **S. Lee**, "Making MambaVision Temporal: Leveraging TSM for Efficient Video Understanding," in *Proc. of The 24th International Conference on Electronics, Information, and Communication (ICEIC 2025)*, Osaka, Japan, Jan, 2025.
- **S. Lee**, S. Hong and E. Kim, "D-TSM: Discriminative Temporal Shift Module for Action Recognition," in *Proc. of The 20th International Conference on Ubiquitous Robots (UR 2023)*, Hawaii, USA, Jun, 2023.
- **S. Lee**, J. Kim, S. Hong, B. Kim, D. Noh and E. Kim, "Semantic Segmentation Using Superpixel and Fully Convolutional Network," in *Proc. of The 32th Institute of Control, Robotics and Systems Annual Conference (ICROS 2017)*, Sokcho, Korea, May, 2017, pp. 31-32.
- **S. Lee**, S. Kim and E. Kim, "Control of Camera Optical Image Stabilization System using Command Shaping Techniques and Feedback Controller," in *Proc. of Conference on Information and Control Systems (CICS 2016)*, Danyang, Korea, Oct, 2016, pp. 13-14.
- **S. Lee**, H. Eum, S. Jang, M. Park and C. Yoon, "Design of Fuzzy Controller for Thermoelectric Cooling System using Condensed Water," in *Proc. of Korean Institute of Intelligent Systems Spring Conference 2015 (KIIS 2015)*, Ansan, Korea, Apr, 2015, pp. 159-160.
- **S. Lee**, S. Jang, M. Park and C. Yoon, "Cooling System Control Based on Fuzzy Look-Up Table Using Temperature Sensor," in *Proc. of Korean Institute of Intelligent Systems Fall Conference 2014 (KIIS 2014)*, Gangneung, Korea, Oct, 2014, pp. 70-71.
- **S. Lee**, J. Baek and M. Park, "Design of Dynamic Hysteresis Model based on T-S Fuzzy Duhem Model," in *Proc. of The 14th International Symposium on Advanced Intelligent Systems (ISIS 2013)*, Daejeon, Korea, Nov, 2013.

Patents Applications

- "ROBOT AND METHOD FOR CONTROLLING THEREOF", **KR** 10-2020-0172983, Dec, 2020.
- "METHOD OF ASSISTING AUTONOMOUS VEHICLE AND APPARATUS THEREFOR", **KR** 10-2018-0094612, **CN** 201980054390, **EP** 19850615, **US** 201916539312, Aug, 2018.

Academic activities

Reviewer

- International Journal of Fuzzy Logic and Intelligent Systems
- The Journal of Korean Institute of Next Generation Computing
- Journal of Institute of Control, Robotics and Systems

- The Journal of Supercomputing
- International Journal of Control, Automation, and Systems
- Journal of Big Data

Honors and Awards

- 2023 Performance Evaluation Award, Sungkonghoe University
- 2022 Performance Evaluation Award, Sungkonghoe University